

C-SQD Platform Specification

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Part I

Epistemic Auditing Infrastructure

1 Introduction

Knowledge-dependent institutions — in science, medicine, policy, and industry — rely on the evaluation of claims. A clinical trial must be assessed before a drug reaches patients. A scientific finding must be scrutinized before it shapes further research. A model or benchmark must be interrogated before it informs consequential decisions. Yet the infrastructure that supports this evaluative work is fragmented, poorly compensated, and largely invisible.

C-SQD is infrastructure for **epistemic auditing**: the structured evaluation of claims, evidence, datasets, models, protocols, and conclusions by qualified reviewers operating under transparent rules and incentives. The platform’s central premise is that expert scrutiny is among the most valuable activities in any knowledge-intensive domain, yet it remains underfunded, weakly recognized, and scattered across institutions.

C-SQD transforms epistemic auditing into a first-class scholarly and economic activity. The platform provides the infrastructure through which expert evaluation can be requested, funded, performed, recognized, and archived — across domains.

1.1 The Problem of Invisible Scrutiny

Modern knowledge production depends on review. Scientific journals rely on peer review. Regulatory agencies rely on expert evaluation. Companies rely on internal and external validation. Researchers rely on informal feedback and methodological critique.

Despite its importance, much of this work remains invisible. High-quality review labor is rarely compensated, often difficult to discover, and only weakly incorporated into professional evaluation. The result is a system that depends on scrutiny while providing limited infrastructure for supporting it.

This problem is not unique to academic publishing. It is structural: any domain that produces claims — and relies on those claims to make decisions — faces the same misalignment between the value of rigorous evaluation and the recognition and resources it receives.

1.2 Intellectual Foundations

Mediating Assessments Protocol

The platform’s design draws inspiration from Kahneman, Sibony, and Sunstein’s Mediating Assessments Protocol (MAP), a framework introduced to reduce noise in judgment. Although bias in evaluation receives considerable attention, noise — unwanted variability in assessments — remains underaddressed across domains. MAP principles inform several C-SQD design choices:

- Replacing binary accept/reject verdicts with structured, ongoing evaluation
- Using multidimensional evaluation tuples (see Section 3.2) rather than single aggregate metrics
- Employing independent assessments (*ElementReviews*) along distinct dimensions of artifact quality

Conjecture and Criticism

Science and other knowledge-intensive fields advance through conjecture and rigorous criticism. Yet existing institutional structures reward conjecture — novel claims, positive findings, exciting results — while systematically undervaluing criticism. C-SQD’s goal is to correct this imbalance by creating incentive structures that recognize and compensate the evaluative work on which all knowledge ultimately depends.

2 Core Concepts

2.1 The Audit Object

An **audit object** is any artifact containing claims that can be subjected to structured evaluation. Depending on the domain, audit objects may include:

- Research manuscripts, preprints, and datasets
- Clinical trial protocols and regulatory submissions
- AI model benchmark specifications and evaluation reports
- Policy proposals and technical standards
- Software systems and algorithmic decision tools

The platform treats audit objects uniformly at the infrastructure level. Domain instantiations define which object types are supported and what evaluation criteria apply to them.

2.2 The Common Weakness Enumeration

The **Common Weakness Enumeration** (CWE) is a structured taxonomy of evaluation criteria applicable to audit objects within a given domain. The CWE is the primary substrate against which ElementReviews are performed.

Each domain instantiation of C-SQD maintains its own CWE, potentially extending a shared base taxonomy. Communities of reviewers may supplement the base CWE with field-specific criteria (see Section 3.3). Examples:

- In academic publishing, the CWE (denoted CRWE, Common Research Weakness Enumeration) covers experimental design, causal & statistical methodology, reproducibility, dataset quality, etc.
- In clinical trial review, the CWE covers protocol completeness, endpoint validity, causality, statistical power, blinding integrity, regulatory alignment, etc.
- In AI auditing, the CWE covers benchmark construction, interventional competence, distributional shift, evaluation validity, documentation completeness, etc.

2.3 Review Mechanisms

C-SQD implements two core review mechanisms that apply across all domain instantiations.

ElementReviews (ERs) ElementReviews are focused evaluations that assess an audit object against a specific criterion from the CWE. The platform algorithmically generates ElementReview Solicitations (ER-S) that match reviewers with audit objects, specific CWE criteria, and a payment scheme. Matching is based on demonstrated reviewer expertise with respect to the CWE criterion and the subject matter of the audit object. Reviewers may also submit unsolicited ElementReviews (unpaid).

ElementReviews ensure a thorough multidimensional assessment while maintaining appropriate anonymity during review. All ElementReviews are visible on the platform, but reviewer identities are revealed only after a configurable delay following the close of the initial review phase.

SynthesisReviews SynthesisReviews are integrative evaluations analogous to traditional narrative reviews. After an initial public review period during which ElementReviews accumulate, the platform may solicit SynthesisReviews that integrate and contextualize those ElementReviews. This separation of broad initial review and expert synthesis combines crowd wisdom with structured expert evaluation.

SynthesisReviews are unpaid but confer high status. They are solicited only for audit objects that reach a significance threshold based on their evaluation tuple at the close of the public review phase.

3 Platform Mechanics

3.1 Review Phases

The platform implements a three-phase review process. Specific phase parameters — durations, anonymity rules, payment structures — are configured per domain instantiation. The phases may partially overlap but are meant to be roughly sequential.

Phase 1: Public Review The public review period is open to all qualified platform users. Only solicited reviews may receive compensation. The C-SQD algorithm creates ER-S assignments stochastically; each ER-S specifies a reviewer, an audit object, a CWE criterion, and a payment scheme. Users who do not initially receive an ER-S may petition to be solicited.

The public review period continues until allocated review funds are depleted. After closing, reviewers may still submit unpaid reviews and accrue status benefits. Reviewers or external entities may fund additional review rounds or establish challenge bounties.

Phase 2: Submitter Response During and after the public review period, the submitter of the audit object may respond to reviews, accept or contest findings, and submit revisions. Domain instantiations define the number of permitted revision rounds and any conditions under which revisions trigger a new review phase. At the end of Phase 2, the evaluation tuple is finalized (see Section 3.2).

Phase 3: Expert Synthesis Review Audit objects that reach a significance threshold may be solicited for expert SynthesisReview. During this phase, submitter and reviewer identities are maintained under the anonymity rules configured for Phase 1. The expert reviewer and the submitter remain anonymous to each other if Phase 1 was double-blind. SynthesisReviews are unpaid but carry high status and become part of the reviewer’s permanent public record.

3.2 Audit Object Evaluation

The platform evaluates audit objects through a tuple function:

$$E(A \mid R, T_{\text{eval}}) \rightarrow (N, M, S, L, U)$$

This function takes as input an audit object A (including supplementary materials such as code, data, or protocols), a specific reviewer community R , and a reference time T_{eval} . It outputs a five-dimensional evaluation tuple where:

- N measures reported non-ethical problems identified by reviewers (higher values indicate more issues)
- M measures reported ethical problems identified by reviewers (higher values indicate more issues)
- S quantifies the *stakes* of the audit object — its consequentiality for downstream decisions, practice, or policy. Domain instantiations configure how S is operationalized: in academic publishing, S reflects the significance of a scientific contribution; in clinical trial review, S reflects the potential public health impact of the trial; in AI auditing, S reflects the deployment scope and risk profile of the system under review.
- L represents the depth of review scrutiny the audit object has received, reflecting both the volume and the expertise-weight of ElementReviews
- U tracks *uptake* — the degree to which the audit object, and the audit record associated with it, has propagated into subsequent artifacts or decisions. Domain instantiations configure how U is measured: citations in academic publishing; guideline adoptions or regulatory references in clinical contexts; deployment decisions or downstream model evaluations in AI auditing.

This multidimensional approach preserves important distinctions that a single metric would collapse. For example, an audit object may carry substantial problems identified in review (N high) while remaining highly consequential (S high) — a distinction that matters differently to different consumers of the evaluation. Submitters can influence L directly by paying for higher-tier review, or indirectly by producing work that attracts unsolicited reviews and high challenge rates.

Platform users can dynamically recompute evaluations for different reviewer communities and set default reviewer criteria for their interface. This flexibility serves diverse needs: a drug regulator might recompute E using only biostatistician and clinical pharmacologist communities; a policy analyst might weight S and U over N when assessing whether a contested finding has nonetheless shaped practice; an AI safety team might filter by reviewers with demonstrated expertise in a specific CWE criterion when auditing a model evaluation report.

3.3 The Reviewer Network

Communities and Tags

The platform allows reviewers to form specialized communities based on shared expertise, methodology, or institutional affiliation. C-SQD implements communities through a tag system. Tags identify reviewer attributes and can be combined to define community lenses through which audit object evaluations are recomputed.

1. Users can filter reviewers by combining any tags (e.g., “neuroscience” + “electrophysiology”).
2. The platform recalculates evaluation tuples using only the selected reviewer group.

Communities emerge naturally as tags gain adoption rather than through formal creation processes. Platform users may establish *verified tags* (v-tags) with defined membership criteria and an associated platform fee.

Professional Relationships

The platform tracks two types of professional connections:

1. **Collaboration networks:** Form automatically when reviewers jointly author reviews.
2. **Lineage networks:** Form when reviewers establish mentor–mentee relationships that are mutually confirmed.

3.4 Reviewer Status and Evaluation

During an initial grace period, requirements for reviewer status are relaxed to facilitate platform growth. After this period, reviewers must meet one of two criteria: receive an endorsement from a reviewer who has completed at least one expert SynthesisReview, or receive x endorsements from reviewers who have each completed at least y ElementReview solicitations.

All endorsements specify the domains and fields to which they apply and become public information on both parties’ profiles.

Reviewers who accept ER-S assignments are expected to submit within three weeks or notify the platform of inability to complete. Moderate algorithmic penalties apply for submissions between three and five weeks; more severe penalties apply beyond six weeks or for violations of the ethical code of conduct.

3.5 The Challenge System

The challenge system addresses two problems inherent to large-scale review:

- With many reviews accumulating, it is valuable to maintain a featured subset representing the best current evaluation of an audit object.

- Beyond compensating reviewers for initial contributions, the platform needs incentives for ongoing adversarial scrutiny of competing critiques.

Reviewers may initiate either:

- **Direct Challenges:** Challenge an existing featured `ElementReview` or `SynthesisReview`. If the challenger’s review wins, it replaces the incumbent in the featured subset.
- **Petition Challenges:** Publicly petition for another reviewer’s review to replace an existing featured review of the same type.

Initiating a direct challenge incurs a modest fee to discourage frivolous challenges and sustain platform finances. Petition challenges incur no fee.

Challenge Rules

- All challengers are identified.
- Challengers must set an election date at least one week in the future, not overlapping with existing challenges for the same featured review.
- Voters may change votes freely; votes at the time of the election determine the winner.
- Each vote is weighted by alignment between the voter’s review record and the featured review in terms of demonstrated expertise.
- Former incumbents cannot rechallenge within 30 days.
- If the incumbent previously won an election, votes from that election default to them in their next election.
- There may be at most m challenges to a given review within any 30-day window.

3.6 Funding Mechanics

Audit pool funding and audit pool allocation are distinct concerns. The former addresses where review compensation comes from; the latter addresses how those funds are distributed across audit objects once pooled. This section treats each in turn.

3.6.1 Audit Pool Funding Sources

Audit pools may be capitalized through three mechanisms, which are not mutually exclusive. A single audit object may draw from more than one source simultaneously, subject to the per-object cap described in Section 3.6.2.

Submission Fees. Submitters pay a fee at the time of submission. Submission fees are tiered, with a minimum of \$0. Higher tiers contribute more to the submitter’s dedicated pool allocation and unlock corresponding benefits (see Section 3.7). The free tier contributes nothing to the dedicated pool; the audit object remains discoverable and eligible for unsolicited community review, but receives no funded ER-S assignments from its own submission.

Illustrative tier structure (specific values are deployment-configurable):

- **Tier 0 (\$0):** Object is publicly discoverable; eligible for unsolicited community review; no dedicated pool allocation; no platform promotion; no reviewer customization.
- **Tier 1:** Small dedicated pool allocation; basic platform visibility; standard ER-S matching.
- **Tier 2:** Larger dedicated pool allocation; enhanced feed visibility; reviewer community preferences honored; eligibility for one revision round with targeted re-review.
- **Tier 3+:** Largest dedicated pool allocation; maximum visibility; full reviewer customization; extended ongoing review support; access to social and reputational markers.

Sponsored Programs. An organization — a funder, regulator, journal, conference, or other institutional actor — establishes a pool covering a defined set of audit objects. Sponsored programs may target a subject area, a submission window, a specific CWE criterion, or a named set of audit objects. Submission fees may be waived or reduced for objects covered by a sponsored program. The Conference Partnership Program (Section 4) is one instance of this mechanism.

Institutional Subscriptions. A portion of institutional and enterprise subscription revenue flows into a domain-level flexible pool. This pool supports review of audit objects that lack a dedicated submission-fee allocation or sponsored program — ensuring that high-stakes objects can receive funded scrutiny regardless of whether their submitter paid. This is the mechanism most directly aligned with the platform’s public-interest mandate: important work should not go unreviewed simply because its submitter could not or did not pay.

3.6.2 Pool Allocation Mechanics

Regardless of funding source, pooled review funds are allocated through the following hybrid model. Specific parameters are configured per domain instantiation.

1. **Guaranteed Minimum Allocation.** For audit objects with a non-zero dedicated pool (from submission fees or a sponsored program), a fixed percentage of that allocation (e.g., 30–50%) is reserved exclusively for compensating reviewers who submit valid `ElementReviews` on that object. This ensures that every funded submission receives at least a baseline level of paid review.
2. **Flexible Pool.** The remaining portion of all funding sources — the non-reserved share of submission fees, the institutional subscription contribution, and any unused dedicated allocations that expire unredeemed — flows into a central flexible pool. This pool is managed by an algorithm that weighs submission volume, reviewer availability, and community engagement signals to direct resources where they are most needed.
3. **Engagement-Triggered Allocations.** Audit objects that attract high community engagement — ER-S petitions, multiple challenges, bug bounty proposals — while receiving disproportionately low unsolicited review are allocated additional funds from the flexible pool. This directs attention toward objects the community deems especially important or contested, independent of what their submitter paid.
4. **Per-Object Cap.** To prevent a small number of prominent submissions from monopolizing the flexible pool, total funds drawn from all pool sources for any single audit object are capped:

$$\text{Total pool funds allocated to any audit object} \leq k \times B,$$

where B is the object’s baseline allocation (defined as a deployment-configurable floor for Tier 0 objects, and as the dedicated allocation for higher-tier objects), and k is initially set (e.g., $k = 2$ or 3) and adjusted based on platform data and community feedback. This formulation ensures the cap scales with contribution while still permitting Tier 0 objects of genuine community importance to receive some flexible pool support.

3.7 Submission Tier Benefits

Higher submission tiers unlock benefits across four categories. These benefits are the primary mechanism by which submitters are incentivized to contribute to review funding beyond the \$0 minimum.

Visibility. The platform provides a customizable feed that surfaces new and significant audit objects. Feed ranking depends on the evaluation tuple, giving submitters an incentive to improve their depth score L by paying for review.

Review Customization. Submitters who pay the review fee may petition for additional or specialized CWE criteria, choose preferred reviewer communities, and request methodological or domain-specific validation.

Ongoing Support. Paying submitters may receive extended review support, partial fee waivers for challenges, and mechanisms for integrating major corrections or addenda.

Social and Reputational Incentives. Paid review may confer visible markers of commitment — distinctive badges, elevated status tiers, or access to exclusive networking events — that build social capital and professional prestige.

3.8 AI Utilization

Platform Tools

AI can help the platform understand which problems its users are interested in, improving:

- **Search:** Problem-based audit object discovery as a more useful alternative to keyword search.
- **Feed:** Personalized discovery algorithms for audit objects and reviews.

A fine-tuned model trained on platform data — audit objects, reviews, supplementary materials — can help users understand prior work, identify non-obvious connections across subfields, and solve practical research problems.

User Tools

AI tools are research products and should themselves be subject to ongoing peer review. Reviewers may use sufficiently peer-reviewed AI tools to supplement their evaluations, subject to disclosure requirements and justification of the tool’s suitability to the specific audit object. Community-developed AI packages may be made available to other users for a fee payable to the creator.

3.9 Revenue Streams

Potential revenue sources include: institutional subscriptions; enterprise subscriptions; platform fees on review pools; sponsored review programs; premium analytics; APIs; workflow partnerships; professional memberships; v-tag fees; challenge fees; bug bounty fees; and nonstandard evaluation requests.

Financial sustainability derives not primarily from submission fees — which largely fund review — but from secondary streams associated with the social network and AI tools.

Part II

Domain Instantiations

4 Instantiation I: Academic Publishing

4.1 Context and Motivation

The current academic publishing system undermines scientific progress. Science advances through conjecture and rigorous criticism, yet traditional publishing models fail to reward the latter. This misalignment creates systemic problems at multiple levels of scientific inquiry.

Academic journals profit from publishing exciting but often unreliable results, while scientists face pressure to produce positive findings or risk their careers. The critical work of peer review remains untransparent, uncompensated, and unrecognized in academic careers. This creates several fundamental problems:

1. **Misaligned Incentives:** Traditional journals profit from restricting access rather than improving reliability.
2. **Publication Bias:** The system incentivizes positive results over negative findings, regardless of scientific value.
3. **Devalued Criticism:** Peer review, essential to scientific progress, lacks recognition and compensation.
4. **Limited Transparency:** Review processes remain opaque, preventing community participation in critical evaluation.

Academic publishing represents the most lucrative segment of the global publishing industry. The top five conglomerates control approximately 50% of market share, generating roughly \$2 billion in annual profit at 20–40% margins. The issue is not the magnitude of these profits but that they derive from monetizing academics’ free labor while restricting knowledge access rather than enhancing reliability.

4.2 Configuration

Audit Object

Research manuscripts, including supplementary materials such as datasets and code.

Common Research Weakness Enumeration (CRWE)

The academic instantiation uses the CRWE, which covers: experimental design; statistical methodology; causal reasoning; reproducibility; dataset quality; benchmark construction; model behavior; protocol design; and evidentiary support. Reviewer communities may extend the CRWE with field-specific criteria.

Evaluation Tuple Configuration

The general evaluation tuple $E(A \mid R, T_{\text{eval}}) \rightarrow (N, M, S, L, U)$ is operationalized in the academic instantiation as follows:

- S (stakes) is operationalized as *scientific significance*: the degree to which the manuscript makes a novel, consequential contribution to its field, assessed by reviewers during SynthesisReview and via community engagement signals.
- U (uptake) is operationalized as *citation impact*: the degree to which the manuscript and its audit record have been cited or built upon in subsequent work, tracked via platform and external citation data.

Review Phase Parameters

Phase 1: Public Review Two variants are offered:

- **Double-Blind Review (author pays review fee):** Both authors and reviewers remain anonymous throughout. The algorithm creates ER-S assignments stochastically, matching reviewers to manuscripts and CRWE criteria based on expertise and subscription status. The public review period continues until allocated funds are depleted. ElementReviews are reidentified and enter reviewers' public records after all three phases complete, with a configurable delay.
- **Non-Anonymous Review (author does not pay review fee):** Authors who do not pay may still publish on the platform. These manuscripts enter a public review period with a maximum duration (e.g., two years). All review submissions are uncompensated. The platform does not advertise the manuscript, but it remains discoverable.

Phase 2: Revisions and Author Response Authors may submit a response with revisions up to two times, accepting or contesting findings from Phase 1. Additional analyses or experiments may warrant a second public review period and additional fees. The evaluation tuple is available at the end of Phase 2.

Phase 3: Expert Synthesis Review Expert SynthesisReviews are solicited for manuscripts that reach a significance threshold based on the evaluation tuple after Phase 1. Author and expert reviewer identities are maintained under the same anonymity rules as Phase 1. Authors may specify preferences for reviewer communities, which bias but do not constrain algorithmic selection. SynthesisReviews are unpaid but carry high status.

Reviewer Concurrency Limits

Each reviewer may hold at most $N_{\text{pub-max}}$ concurrent active ElementReviews during Phase 1, and at most $N_{\text{expert-max}}$ concurrent active SynthesisReviews during Phase 3.

4.3 Platform Components

The Referee Campaign

Referee serves as the entry point for manuscripts first presented elsewhere — on preprint servers or in traditional journals. These manuscripts can be uploaded, discovered, and reviewed on the C-SQD platform.

4.4 Growth Strategy

Strategic Network Engagement

Scientists collaborate within small, overlapping networks formed through shared academic experiences. C-SQD accelerates adoption by partnering with principal investigators (PIs), offering small discretionary grants to their labs in exchange for participation. Each PI commits to:

- Publishing at least one modular manuscript on C-SQD for each submission to a traditional journal
- Citing this modular manuscript in their traditional journal submission
- Providing a minimal level of peer review on the platform

Conference Partnership Program

The Conference Partnership Program streamlines publication and review for conference-associated manuscripts. Conference organizers pay a single fee for waived platform fees across all manuscripts published under their conference banner. Upon enrollment:

- Attendees add a conference-specific tag to their reviewer profiles

- This tag restricts reviewer matching to eligible participants
- The platform manages reviewer–manuscript matching and ER-S assignment
- Organizers determine anonymity levels; all reviews become public after the conference

Conference manuscripts receive DOIs equivalent to any other C-SQD publication. All submitted reviews enter reviewers’ public records.

5 Instantiation II: Clinical Trial Review

5.1 Context and Motivation

Clinical trials are among the most consequential knowledge-producing activities in medicine, yet the processes by which they are designed, evaluated, and approved remain fragmented and inconsistently rigorous. The same structural problems that afflict academic publishing are present here, often with higher stakes: underpowered studies, outcome switching, selective reporting, and opaque regulatory review can result in treatments entering clinical practice without adequate evidentiary support — or in effective treatments being delayed.

The current landscape involves multiple distinct evaluative actors — institutional review boards (IRBs), data and safety monitoring boards (DSMBs), regulatory agencies, journal peer reviewers, and post-market surveillance bodies — each operating under different rules, with limited coordination and almost no public visibility into the quality of evaluation performed.

C-SQD Clinical provides shared infrastructure for epistemic auditing across the clinical trial lifecycle, from protocol design through results dissemination.

5.2 Configuration

Audit Objects

Clinical trial protocols; interim and final results reports; regulatory submissions; systematic reviews and meta-analyses; and post-market surveillance reports.

Clinical Weakness Enumeration (CWE-C)

The clinical instantiation uses CWE-C, which covers:

- **Protocol design:** Endpoint validity, eligibility criteria, randomization and allocation concealment, blinding integrity, sample size and power calculations
- **Statistical methodology:** Pre-registration alignment, multiple comparisons adjustment, handling of missing data, subgroup analysis discipline
- **Operational integrity:** Protocol deviations, adverse event reporting completeness, data monitoring procedures
- **Regulatory alignment:** ICH guideline compliance, informed consent adequacy, data sharing obligations
- **Reporting quality:** CONSORT/SPIRIT adherence, outcome switching, selective reporting of results

Specialized reviewer communities (e.g., oncology trial methodologists, biostatisticians, patient advocates) may extend CWE-C with indication-specific or methodology-specific criteria.

Evaluation Tuple Configuration

The general evaluation tuple $E(A \mid R, T_{\text{eval}}) \rightarrow (N, M, S, L, U)$ is operationalized in the clinical instantiation as follows:

- S (stakes) is operationalized as *public health consequentiality*: an assessment of the potential population-level impact of the trial’s intervention, weighting factors such as indication prevalence, unmet medical need, and the degree to which the trial result could alter clinical practice or regulatory status.
- U (uptake) is operationalized as *downstream adoption*: the degree to which the trial protocol or results have been referenced in subsequent regulatory submissions, clinical guidelines, systematic reviews, or other registered trials, tracked via platform and external registry data.

Review Phase Parameters

Phase 1: Protocol Review Protocol-stage review is the highest-leverage point for epistemic auditing in clinical research. The platform supports two Phase 1 configurations:

- **Sponsor-funded review:** The trial sponsor pays a review fee. ER-S assignments are algorithmically generated against CWE-C criteria and matched to reviewers with demonstrated expertise in the relevant indication and methodology. Anonymity is configurable; sponsor identity is always disclosed, but principal investigator identity may be blinded.
- **Registry-linked review:** Protocols registered in public trial registries (e.g., ClinicalTrials.gov, EudraCT) can be uploaded and subjected to uncompensated community review. This variant is analogous to the non-anonymous academic track — no platform advertising, but the protocol remains discoverable and reviewable.

Phase 2: Sponsor Response and Protocol Amendment Sponsors may respond to Phase 1 reviews and submit protocol amendments. The platform tracks amendment history against the original protocol, creating a permanent record of how the design evolved in response to scrutiny. Amendments that constitute substantive changes to primary endpoints or statistical analysis plans trigger a second round of targeted ER-S assignments.

Phase 3: Results Synthesis Upon completion of the trial, the results report is linked to the Phase 1 protocol audit record. Expert SynthesisReviews are solicited that evaluate the results in light of the pre-registered protocol and the Phase 1 review record. This creates a direct audit trail from design scrutiny to results interpretation, making outcome switching and post-hoc analysis detectable and part of the permanent public record.

Anonymity and Disclosure

Clinical trial review involves real conflicts of interest — between sponsors, investigators, and the public — that require careful disclosure management. The platform’s identity and conflict disclosure infrastructure (Section 2) is configured to require:

- Mandatory disclosure of any financial relationship between reviewer and sponsor
- Mandatory disclosure of any prior advisory or consulting relationship with the sponsor
- Exclusion of reviewers with undisclosed material conflicts

5.3 Reviewer Communities

The clinical instantiation supports specialized reviewer communities including: trial methodologists; biostatisticians; indication-specific clinicians; patient and public involvement (PPI) advocates; regulatory scientists; and health economists. Each community may extend CWE-C and filter evaluation tuples through their community lens.

Patient and public involvement communities occupy a distinctive role: they provide evaluation dimensions — relevance of endpoints to patients, burden of participation, adequacy of informed consent — that purely technical reviewer communities may underweight.

5.4 Integration with Existing Infrastructure

C-SQD Clinical is designed to complement rather than replace existing regulatory and oversight mechanisms. The platform’s public audit record can serve as:

- A resource for IRBs and DSMBs reviewing protocols
- An input to regulatory submissions (audit depth score L and evaluation tuple E as supplementary documentation)
- A public accountability layer for post-market surveillance
- An evidence base for systematic reviewers and guideline developers

The platform does not adjudicate regulatory decisions. It provides structured, transparent, publicly accessible scrutiny that informs those decisions.

5.5 Growth Strategy

Anchor Institutions

The clinical instantiation is most efficiently adopted through anchor institutions — academic medical centers, cooperative oncology groups, and large pragmatic trial networks — whose protocols represent both high stakes and high public interest. The platform will partner with a small number of such institutions to establish initial review pools and build reviewer community depth in key indications.

Registry Integration

A partnership with public trial registries provides a pipeline of protocols for community-level review even without sponsor participation, building reviewer expertise and platform credibility in advance of sponsor adoption.

Funder Requirements

Major public and philanthropic trial funders (NIH, NIHR, Wellcome Trust) are natural institutional subscribers, both because they benefit from improved protocol quality among their funded portfolios and because mandating C-SQD review as a funding condition would accelerate adoption across the field.

Part III

Platform Governance and Long-Term Vision

6 Governance and Independence

The credibility of the platform depends on the independence of its audit processes. Submitters and sponsors may define scope, eligibility requirements, confidentiality windows, and reward levels. They do not control conclusions.

C-SQD manages reviewer qualification, conflict disclosure, audit procedures, and publication policies across all domain instantiations. Compensation is intended to influence participation, not findings. The objective is a system in which incentives support scrutiny without compromising integrity.

7 The Reputation Economy

A central objective of C-SQD is to create durable value around expert scrutiny. Today, a reviewer may spend years evaluating manuscripts, auditing protocols, or reproducing analyses without generating a persistent public record of those contributions.

C-SQD converts these activities into visible scholarly and professional artifacts. Reviews become attributable, discoverable, citable, measurable, and portable. Reviewers accumulate evidence of expertise that supports hiring, promotion, consulting, collaboration, and professional recognition across domains.

8 Positive Feedback Loops

Three interlocking economies reinforce one another:

- **The infrastructure economy:** Institutional subscriptions support identity systems, governance, analytics, and platform operations. A portion of subscription revenue may support a public Review Fund enabling audits of high-importance artifacts even without a dedicated sponsor.
- **The audit economy:** Organizational demand for credible scrutiny generates funded review pools. Market signals direct resources toward artifacts that participants believe deserve evaluation.
- **The reputation economy:** High-quality scrutiny generates reputation. Strong reviewer reputations increase the value of organizational participation. Increased participation generates additional audit opportunities.

Over time, these reinforcing loops strengthen the entire ecosystem independently of any particular domain instantiation.

9 Long-Term Vision

C-SQD creates a market for epistemic auditing. Organizations obtain credible scrutiny. Experts receive compensation and recognition. Institutions strengthen quality control and researcher development. Readers and decision-makers gain access to structured evaluations of important claims.

The long-term opportunity extends beyond academic publishing and clinical trials. Any domain that depends on trustworthy evaluation of evidence can benefit from structured epistemic auditing infrastructure. As the network expands across domains, C-SQD becomes a general-purpose layer for trustworthy evaluation across science, medicine, artificial intelligence, policy, and other knowledge-intensive fields.